RDocumentation

Learn R

Search all packages and functions

ResourceSelection (version 0.3-6)

hoslem.test: Hosmer-Lemeshow Goodness of Fit (GOF) Test

Description

Hosmer-Lemeshow Goodness of Fit (GOF) Test.

Usage

```
hoslem.test(x, y, g = 10)
```

Value

A list with class `"htest"` containing the following components:

`statistic`	the value of the chi-squared test statistic, (`sum((observed - expected)^2 / expected)`).
`parameter`	the degrees of freedom of the approximate chi-squared distribution of the test statistic (`g - 2`).
`p.value`	the p-value for the test.
`method`	a character string indicating the type of test performed.
`data.name`	a character string giving the name(s) of the data.
`observed`	the observed frequencies in a `g`-by-2 contingency table.
`expected`	the expected frequencies in a `g`-by-2 contingency table.

Arguments

x	a numeric vector of observations, binary (0/1).
y	expected values.
g	number of bins to use to calculate quantiles. Needs to be at least 2. The degrees of freedom of the test is `g-2` but the number of bins might also depend on the data (i.e. due to identical quantile values that lead to empty bins). In case when the number of bins does not equal `g`, the degrees of freedom will depend on the number of bins that is less than `g`. In such case a warning is produced.

Author

Peter Solymos by adapting code pieces from R help mailing list

Details

The Hosmer-Lemeshow test is a statistical test for goodness of fit for logistic regression models.

References

Hosmer D W, Lemeshow S 2000. Applied Logistic Regression. New York, USA: John Wiley and Sons.

Examples

Run this code

```
set.seed(123)
n <- 500
x <- rnorm(n)
y <- rbinom(n, 1, plogis(0.1 + 0.5*x))
m <- glm(y ~ x, family=binomial)
hoslem.test(m$y, fitted(m))
```

Run the code above in your browser using DataLab